# Year 8 SEAL Science
# Energy Practice Test – Easy

| Details | |
|---|---|
| Student Name: | Teacher Name: |
| Date Set: ____/____/26 | Due Date: ____/____/26 |

## Curriculum Assessed

On completion of this unit the student should be able to classify energy as kinetic or potential, calculate kinetic and gravitational potential energy, explain heat transfer by conduction, convection and radiation, analyse energy transfers and transformations using efficiency and Sankey diagrams, calculate appliance running costs, and evaluate ways to reduce energy use. This practice test revises Chapter 7 Energy at an easier level.

## Task Overview

Section A – Multiple Choice: 12 questions, 1 mark each (12 marks). Answer on the separate multiple-choice answer sheet.
Section B – Short Answer: 7 scenario-based questions (28 marks). Answer in this booklet.

## Conditions

- Test conditions
- Formula sheet provided on page 2
- Permitted materials: pens, pencils, calculator, ruler, sharpener, eraser, highlighters.
- Not permitted: notes of any kind, blank paper, white-out, mobile phones, smart watches or any other electronic device.
- 55 mins writing time

## Marking

- 40 marks
- Marked according to teacher-made marking guide.
- Show all working in calculation questions; give every final answer with a unit. When you explain, use the numbers you are given.

| A+ | A | B+ | B | C+ | C | D+ | D | E+ | E | NS |
|---|---|---|---|---|---|---|---|---|---|---|
| 90-100% | 80-89% | 75-79% | 70-74% | 65-69% | 60-64% | 55-59% | 50-54% | 45-49% | 40-44% | <40% |

| Total Marks: | | Percentage: | | Letter Grade: | |
|---|---|---|---|---|---|

## Formula sheet

Use these formulas and units in Sections A and B.

| Quantity | Formula |
|---|---|
| Kinetic energy | KE = ½ × $m$ × $v^2$ ($m$ in kg, $v$ in m/s, KE in J) |
| Gravitational potential energy | GPE = $m$ × $g$ × $h$ ($g$ = 10 N/kg, $h$ in m, GPE in J) |
| Efficiency | Efficiency = useful energy out ÷ total energy in × 100% |
| Energy used by an appliance | Energy (Wh) = power (W) × time (h) |
| Units | 1 kWh = 1,000 Wh · 1 kW = 1,000 W |
| Running cost | Cost ($) = energy used (kWh) × tariff ($ per kWh) |

## SECTION A – Multiple-choice questions (12 marks)

Answer **all** questions on the separate multiple-choice answer sheet by filling in the letter of the best response. Each question is worth 1 mark. No marks are deducted for incorrect answers. Do **not** write your Section A answers in this booklet.

**Question 1**

Before a morning swim at the beach, Tom eats a banana. In which form is the energy stored in the banana?

**A.** kinetic energy

**B.** chemical potential energy

**C.** thermal energy

**D.** gravitational potential energy

**Question 2**

A 60 kg swimmer climbs the ladder to a diving platform 3 m above the water. How much gravitational potential energy does she gain? (GPE = $m \times g \times h$, $g$ = 10 N/kg)

**A.** 1,800 J

**B.** 180 J

**C.** 600 J

**D.** 900 J

**Question 3**

On a sunny day, dry sand on the beach becomes too hot to walk on. How does energy from the Sun reach the sand?

**A.** by conduction, as air particles pass the heat down to the sand

**B.** by convection, as hot air from the Sun sinks onto the sand

**C.** by radiation, which can travel through empty space

**D.** by friction, as the wind rubs the sand grains together

**Question 4**

On a sunny afternoon, the sand heats up faster than the sea. A cool breeze then blows from the sea onto the beach. Which is the best explanation?

**A.** Cold air from the sea is attracted towards the heat of the sand.

**B.** Heat rises from the sand and pulls the sea air towards the beach.

**C.** Waves breaking on the shore push the sea air onto the beach.

**D.** Warm air above the sand is less dense and rises, and cooler sea air moves in to replace it.

**Question 5**

Each second, the small solar panel on a caravan roof receives 400 J of light energy and supplies 80 J of electrical energy. What happens to most of the other 320 J?

**A.** It is transformed into thermal energy, which warms the panel and the air.

**B.** It is destroyed, because the panel is not perfect.

**C.** It is stored in the panel and given out at night as light.

**D.** It stays trapped as light energy inside the panel's glass.

*Use the following information to answer Question 6.*

A family tests three camping lights. The table shows the energy for each light every second.

**Question 6**

Which camping light is the most efficient?

| Light | Electrical energy in (J) | Light energy out (J) |
|---|---|---|
| P | 4 | 1 |
| Q | 5 | 2 |
| R | 20 | 3 |

**A.** Light P

**B.** Light Q

**C.** Light R

**D.** All three are equally efficient, because energy is conserved.

**Question 7**

Which is one reason First Nations Peoples have used fire to care for Country?

**A.** to make the soil hot enough to plant crops all year round

**B.** because a fire creates new energy from nothing

**C.** to encourage fresh green growth, which attracts animals to the area

**D.** to get rid of all the trees so the land stays open

**Question 8**

Two portable air conditioners for a caravan are the same size. One has a 5-star energy rating and the other has a 2-star energy rating. What do the extra stars tell you about the 5-star model?

**A.** It cools the caravan to a lower temperature.

**B.** It costs more to buy.

**C.** It uses more electricity because it is more powerful.

**D.** It uses less energy to do the same job.

**Question 9**

A 6 W LED strip light in the caravan is on for 5 hours. How much energy does it use, in kWh? (Energy (Wh) = power (W) × time (h); 1 kWh = 1,000 Wh)

**A.** 0.03 kWh

**B.** 30 kWh

**C.** 1.2 kWh

**D.** 0.3 kWh

**Question 10**

A camping esky has thick foam walls. Ice inside it stays frozen for two days on a hot beach. Why does the foam help?

**A.** Foam is a good conductor, so it carries heat away from the ice.

**B.** Foam is an insulator, so it slows heat flowing in from the warm air.

**C.** Foam gives out cold, which flows in and keeps the ice frozen.

**D.** Foam stops all heat from moving, so the ice can never melt.

**Question 11**

On a camping trip, the caravan's solar panel and battery can supply 960 Wh of energy each day. The family's devices need 1,100 Wh each day. Which statement is correct?

**A.** Supply is greater than demand, so they have energy to spare.

**B.** Supply and demand are equal, so the system balances.

**C.** The battery makes extra energy when demand is high, so it balances.

**D.** Demand is greater than supply, so they must use less or add more supply.

**Question 12**

Ruby times how long a bag of ice takes to melt in an esky. She does the test three times and averages the results. Why does she repeat the test?

**A.** so the results are more likely to match her hypothesis

**B.** so that she can change more than one variable at once

**C.** to reduce the effect of random (unpredictable) errors

**D.** so that the thermometer reads more accurately

**END OF SECTION A**

## SECTION B – Short-answer questions (28 marks)

Answer **all** questions in the spaces provided. Show working for all calculations. Give every final answer with a unit.

### Question 1 (6 marks)

**Surf day.** Sam and his surfboard have a total mass of 70 kg. Use $g$ = 10 N/kg.

**a)** For each example, name the type of energy and say whether it is kinetic or potential.

(i) The sound of the waves crashing.

Type: _____ Kinetic or potential: _____

(ii) Sam standing still at the top of a tall sand dune.

Type: _____ Kinetic or potential: _____

**2 marks**

**b)** Sam rides a wave at 4 m/s. Calculate his kinetic energy. Use KE = ½ × $m$ × $v^2$.

**2 marks**

**c)** At the top of the sand dune, Sam's gravitational potential energy is 4,200 J. Calculate the height of the dune. Hint: height = GPE ÷ ($m$ × $g$).

**2 marks**

## Question 2 (4 marks)

**Campfire.** In the evening the family lights a campfire. A metal tent peg lies with one end in the hot coals. Sam hangs his wet socks on a branch high above the fire.

**a)** Identify the main method of heat transfer (conduction, convection or radiation) in each situation.

(i) The tent peg becomes hot all along its length.

Method: _____

(ii) Smoke and hot air rise from the fire and dry the socks.

Method: _____

**2 marks**

**b)** Explain, in terms of particles, how heat travels along the metal tent peg. Start with: *The particles at the hot end …*

**2 marks**

## Question 3 (3 marks)

**Camp stove.** Figure 1 is a Sankey diagram for a gas camp stove heating a pot of water for one second. Arrow widths are drawn to scale.

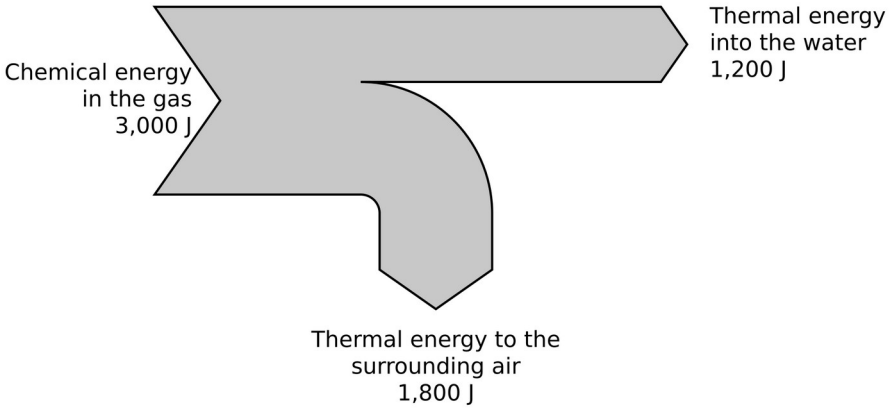



*Figure 1. Sankey diagram for the camp stove (one second).*

**a)** Calculate the efficiency of the camp stove. Use efficiency = useful energy out ÷ total energy in × 100%.

**2 marks**

**b)** On a windy day, the family puts a metal windshield around three sides of the stove. Explain why this makes the stove more efficient.

**1 mark**

## Question 4 (4 marks)

**Electric cooler.** At a powered caravan park site, the family runs an electric cooler. It uses an average power of 50 W, 24 hours a day, for a 10-day trip. Electricity costs $0.30 per kWh.

**a)** Calculate the energy the cooler uses over the 10-day trip. Step 1: find the energy in Wh. Step 2: change it to kWh.

**2 marks**

**b)** Calculate the cost of running the cooler for the trip. Use cost = kWh × tariff ($ per kWh).

**1 mark**

**c)** Instead, the family could fill an ordinary esky with ice. Ice for the same 10 days would cost $25.00. Use your answer to b) to decide which option is cheaper, and by how much.

**1 mark**

## Question 5 (3 marks)

**Fire on Country.** On the trip, the family joins a tour run by a local Traditional Owner. She shows them how to start a fire with a fire saw.

**a)** The steps below are mixed up. Number them 1 to 4 in the correct order.

____ Blow gently on the tinder bundle until it bursts into flame.

____ Saw the edge of one piece of wood quickly across another piece of wood.

____ Put the ember into a tinder bundle of fine, dry grass.

____ A small, hot, glowing ember forms in the wood dust.

**2 marks**

**b)** Name one other method First Nations Peoples use to start a fire.

**1 mark**

## Question 6 (2 marks)

**Hot caravan.** On sunny beach days, the inside of the caravan gets very hot.

Suggest one change to the caravan's design that would keep it cooler inside without using any appliance. Explain how it reduces heat transfer. Start with: *One change is … This keeps the caravan cooler because …*

**2 marks**

## Question 7 (6 marks)

**Painted cans.** Lena wants to find out which colour of can heats up most in sunlight. She paints three identical steel cans matt black, white and shiny silver, and fills each with 250 mL of water at the same starting temperature. She leaves the cans side by side in full sun for 30 minutes, then uses a thermometer to find each temperature rise. She repeats the test on three days.

| Can colour | Rise, trial 1 (°C) | Rise, trial 2 (°C) | Rise, trial 3 (°C) | Average (°C) |
|---|---|---|---|---|
| Matt black | 8.0 | 8.5 | 7.5 | 8.0 |
| White | 3.5 | 4.0 | 3.0 | 3.5 |
| Shiny silver | 1.5 | 5.5 | 2.5 | ? |

**a)** Identify the independent variable in Lena's investigation.

**1 mark**

**b)** Write a hypothesis for Lena's investigation. Use: *If … then … because …*

**1 mark**

**c)** One result for the shiny silver can is an outlier (a result that is very different from the others in its group). Identify it, and state how you can tell.

**1 mark**

**d)** Calculate the average temperature rise for the shiny silver can, leaving out the outlier.

**1 mark**

**e)** Suggest one improvement to Lena's method, and explain how it would make her results more reliable or valid.

**2 marks**

**END OF SECTION B**

## Additional space

Clearly number any response written here.

**END OF TEST**